

Course Code: BCE-O630
Course Name: APPLIED AI

MM: 100 Time: 3 Hr. L T P 3 0 0	Sessional:30 ESE:70 Credit :3
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Prerequisites:	Some exposure to formal languages, logic and programming
Objectives:	This course will help the students to understand the fundamentals of AI techniques. They will learn the concepts of intelligent systems for pattern recognition and statistical reasoning, develop smart applications using AI, understand computational mathematics for learning and data analysis, explore the algorithms for heuristic search and state space search. An introduction to machine learning, deep learning etc. will also be provided.

NOTE:	The question paper shall consist of two sections A and B. Section A contains 10 short type questions of 6 marks each and student shall be required to attempt any five questions. Section B contains 8 long type questions of ten marks each and student shall be required to attempt any four questions. Questions shall be uniformly distributed from the entire syllabus.
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Module	Course Content	No. of Hours	POs mapped	PSOs mapped
<i>Module-1</i>	Introduction of AI: The AI basics, AI Problems, the core assumptions of AI, the need of AI over the traditional methods, the AI Techniques, the level of the Model, Criteria for success and failure of the models, Artificial intelligence fundamentals, Machine learning, Human languages technologies, Distributed systems: paradigms and models for AI, Intelligent systems for pattern recognition, Smart applications using AI, Computational mathematics for learning and data analysis.	08	PO1/ PO2	PSO1/ PSO2
<i>Module-2</i>	Algorithms for Heuristic search and State space search, Defining problem as state space search, Production Systems, Production Characteristics, Production System Characteristics, And issues in the design of search programs, Additional problems. Generate-and-test, Hill climbing, Best-first search, Problem reduction, Constraint satisfaction, Means-ends analysis.	08	PO1/ PO2/ PO3	PSO1/ PSO2
<i>Module-3</i>	Application of AI in Natural language processing, Introduction, syntactic processing, semantic analysis, discourse and pragmatic processing, spell checking, Measures of WordNet similarity, Sentiment analysis and opinions on the web.	08	PO1/ PO2/ PO3/ PO4	PSO1/ PSO2
<i>Module-4</i>	Exploring sub-discipline of AI: Machine Learning, Supervised learning, Unsupervised learning, Reinforcement learning, Classification problems, Regression problems, Clustering problems, Introduction to neural networks and deep learning.	08	PO1/ PO2/ PO3/ PO4	PSO1/ PSO2
<i>Module-5</i>	Statistical Reasoning: Probability and Bays' Theorem, Certainty Factors and Rule-Based Systems, Bayesian Networks, Dempster-Shafer Theory, Fuzzy Logic. AI for knowledge representation, rule-based knowledge representation, procedural and declarative knowledge, Logic programming, Forward and backward reasoning. Introduction to Prolog: Syntax and numeric function, Basic list manipulation functions in Prolog, functions, predicates and conditional, Input, Output and local variables, Iteration and recursion.	08	PO1/ PO2/ PO3/ PO4/ PO12	PSO1/ PSO2
Total No. of Hours		40		

Learning Outcomes:	- Understand the need for Artificial Intelligence and the limitations of conventional methods. - Design, Develop, and Implementation of various AI methods in Prolog, and in Python for prediction, regression, classification, clustering, Searching, and decision-making problems. - Demonstrate the ideas behind selection of various AI methods and their use.
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Suggested books:

S. No.	Name of Authors /Books /Publisher/Year
1.	Nils J. Nilsson, Artificial Intelligence: A New Synthesis (1 ed.), Morgan-Kaufmann, 1998. ISBN 978-1558605350.
2.	Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach (3 ed.), Pearson, 2010. ISBN 978-0136042594.
3.	Bishop, Christopher M, Pattern Recognition and Machine Learning (1 ed.), Springer, 2006. ISBN 978-1493938438.
4.	Daniel Jurafsky and James H. Martin, Speech & language processing (2 ed.), Pearson Education India, 2000. ISBN 978-0131873216.
5.	Shalev-Shwartz, Shai, and Shai Ben-David, Understanding machine learning: From theory to algorithms (3 ed.), Cambridge university press, 2014. ISBN 978-1107512825.

CO-PO/PSO MAPPING														
Course Outcomes (COs)	Program Outcomes (POs)												Program Specific Outcomes (PSOs)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	✓	✓											✓	✓
CO2	✓	✓	✓	✓									✓	✓
CO3	✓	✓	✓	✓								✓	✓	✓